

$$162. \sqrt{.01} \times \sqrt[3]{.008} - .02 = \sqrt{(.1)^2} \times \sqrt[3]{(.2)^3} - .02 \\ = .1 \times .2 - .02 = .02 - .02 = 0.$$

$$163. \text{ Given exp. } = \sqrt[3]{\frac{0.008 + 0.000064}{0.064 + 0.000512}} = \sqrt[3]{\frac{0.008064}{0.064512}} \\ = \sqrt[3]{\frac{8064}{64512}} = \sqrt[3]{\frac{1}{8}} = \frac{1}{2} = 0.5.$$

$$164. \text{ Let } \sqrt[3]{3} = x.$$

$$\text{Then, } (\sqrt[3]{9} - \sqrt[3]{3} + 1) = (x^2 - x + 1) = \frac{x^3 + 1}{x + 1} = \frac{(\sqrt[3]{3})^3 + 1}{(\sqrt[3]{3} + 1)}.$$

$$\Rightarrow (\sqrt[3]{9} - \sqrt[3]{3} + 1)(\sqrt[3]{3} + 1) = (\sqrt[3]{3})^3 + 1 \\ = 3 + 1 = 4, \text{ which is rational.}$$

$$165. \text{ Clearly, } (21)^3 = 9261 \text{ and } (22)^3 = 10648.$$

So, 9261 is the largest four-digit number which is a perfect cube.

$$166. 21600 = 2^5 \times 3^3 \times 5^2.$$

To make it a perfect cube, it must be multiplied by (2×5) , i.e., 10.

$$167. 3600 = 2^3 \times 5^2 \times 3^2 \times 2.$$

To make it a perfect cube, it must be divided by $5^2 \times 3^2 \times 2$, i.e. 450.

$$168. \text{ Required number to be added} \\ = 9^3 - 710 = 729 - 710 = 19.$$

$$169. \begin{array}{r|l} 8 & \overline{79 \ 21} \\ 8 & 64 \\ \hline 169 & 1521 \\ \hline 9 & 1521 \\ & \times \end{array} \quad 89 \\ \Rightarrow \sqrt{7921} = 89$$

$$170. \text{ Given } ? = \sqrt[4]{(625)^3} \\ = (625)^{\frac{3}{4}} = (5 \times 5 \times 5 \times 5)^{\frac{3}{4}} \\ = (5^4)^{\frac{3}{4}} = 5^3 = 125$$

$$171. \text{ Given } \sqrt{y} = 4x$$

$$\Rightarrow y = (4x)^2 = 16x^2$$

$$\Rightarrow y = 16x^2$$

$$\therefore \frac{x^2}{y} = \frac{1}{16}$$

$$172. \text{ Let the number be } x.$$

$$\text{Given: } \sqrt{2025.11} \times \sqrt{256.04} + \sqrt{399.95} \times \sqrt{7} = 33.98 \times 40$$

$$\sqrt{2025} \times \sqrt{256} + \sqrt{400} \times \sqrt{7} = 34 \times 40$$

$$45 \times 16 + 20 \times \sqrt{7} = 1360$$

$$20 \times \sqrt{7} = 1360 - 720 = 640$$

$$\sqrt{7} = \frac{640}{20} = 32$$

$$\Rightarrow ? = (32)^2$$

$$\therefore ? = 32 \times 32 = 1024$$

Hence, the number is 1024.

$$173. \frac{1}{\sqrt{28}} = \frac{1}{\sqrt{2 \times 2 \times 7}} = \frac{1}{2\sqrt{7}} \\ = \frac{\sqrt{7}}{2\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{7}}{14} \\ = \frac{2.645}{14} = 0.189$$

$$174. = \left(\sqrt{\frac{25}{9}} - \sqrt{\frac{64}{81}} \right) \div \sqrt{\frac{16}{324}} \\ = \left(\frac{5}{3} - \frac{8}{9} \right) \div \frac{4}{18}$$

LCM of 3 and 9 is 9.

$$= \left(\frac{15-8}{9} \right) \div \frac{2}{9} = \frac{7}{9} \div \frac{2}{9} \\ = \frac{7}{9} \times \frac{9}{2} = \frac{7}{2} = 3.5$$

$$175. \text{ Given: } 1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$$

$$\Rightarrow 1728 \div \sqrt[3]{64 \times 64 \times 64} \times ?$$

$$= 4491 + 288$$

$$\Rightarrow \frac{1728}{64} \times ? = 4779$$

$$\Rightarrow 27 \times ? = 4779$$

$$\Rightarrow ? = \frac{4779}{27} = 177$$

$$176. \text{ Let the number be } a.$$

$$(\sqrt{7} + 11)^2 \\ = a^{\frac{1}{3}} + 2\sqrt{847} + 122 \\ \Rightarrow 7 + 121 + 22\sqrt{7} \\ = a^{\frac{1}{3}} + 22\sqrt{7} + 122 \\ \Rightarrow 128 - 122 = a^{\frac{1}{3}}$$

$$\Rightarrow a^{\frac{1}{3}} = 6$$

$$\Rightarrow a = (6)^3 = 216$$

Hence, the number is 216.

$$177. \text{ Let the number be } x.$$

$$x - \sqrt{784} = 6 \times \sqrt{324}$$

$$x - 28 = 6 \times 18$$

$$x - 28 = 108$$

$$x = 108 + 28 = 136$$

$$178. \text{ Let the number be } x.$$

$$\text{Given } \sqrt{2116} - \sqrt{1600} = \sqrt{(x)}$$

$$46 - 40 = \sqrt{(x)}$$

$$\Rightarrow 6 = \sqrt{x}$$

$$\Rightarrow x = (6)^2 = 36$$

Hence, the number is 36.